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PERSPECTIVES

40 years of storming: a historical review of Tuckman's model of small group development

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This paper presents a historical overview of the Tuckman model describing the stages of group development. Created by Bruce W. Tuckman in 1965 and revised by Tuckman and Mary Ann Conover Jensen in 1977, the model presents the well-known stages of forming, storming, norming, performing, and adjourning. This model has a unique history in that it was initially popular among HRD practitioners and later became common in academic literature as well. Its significance was a reflection of its time, responding both to the growing importance of groups in the workplace and to the lack of applicable research. This paper uses a literature review to trace the history of the model in terms of field practice and academic research.

Keywords: teams; group development stages; Tuckman, Bruce; HRD history

Human resource development (HRD) is a relatively new academic discipline. Over the past 50 years scholars and practitioners have developed theories and definitions to help us understand who we are, what we are trying to accomplish, and how we can get there. One of the most influential models has been Bruce W. Tuckman's description of the stages of development in small groups. First published in 1965, the model identified the now-classical four stages of forming, storming, norming, and performing.

Tuckman's model has become 'the most predominantly referred to and most widely recognized in organizational literature' (Miller 2003, 122). The model's significance was a reflection of its time, responding both to the growing importance of groups in the workplace and to the lack of applicable research. It proved useful for practice by describing the new ways that people were working together, helping group members understand what was happening in the development process, and providing consultants a way to predict the stages of growth in groups. It proved useful for theory development by providing a common language and what Rickards and Moger (2000, 277) called 'a simple means of discussing and exploring team dynamics'.

This paper chronicles the development of the model, its influence on the community of practice, and its application to academic research. It is based on a

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review of literature, published accounts of the development process, and the author's correspondence with Dr. Tuckman.

Literature review

The literature citing these models is extensive. According to a Google Scholar search conducted by me in July, 2008, Tuckman (1965) was cited in 1196 articles and Tuckman and Jensen (1977) were cited in 544. Given the volume of citations, I attempted to find representative samples of articles to understand how the models were being used in current academic literature.

The first stage of the literature search was to locate articles that specifically supported, refuted, or modified either the Tuckman (1965) or Tuckman and Jensen (1977) models. In addition to Google Scholar's list of articles citing each paper, I used Business Source Premier and the University of Minnesota multi-source search option to find the title keywords *Tuckman* and *group development*. After reviewing the references generated by these searches, I identified four examples. The earliest was Runkel et al. (1971). The other references occurred within the past ten years (Cassidy 2007; Miller 2003; Rickards and Moger 2000).

In addition, I attempted to create a snapshot of the extent to which the model is being used in business literature. A search on Business Source Premier for the keyword 'team development' yielded 27 articles on the development of teams and work groups. Of these, 22% cited at least one version of the Tuckman model.

Finally, I conducted an information search about B.W. Tuckman, including visits to his Ohio State University home page (Tuckman 2003) and online curriculum vitae (Tuckman 2008). Other sources included Tuckman's published reflections on the creation of the model (1984) and influences on his professional career (1996). I also forwarded a final draft of this article to Professor Tuckman, who graciously read it, suggested improvements, and confirmed that it 'accurately described my work and what is going on in the field of group development' (personal communication, 28 July 2008).

Development of the model

Background of Bruce W. Tuckman

Dr. Bruce W. Tuckman (1938–) is an educational psychologist and researcher. His career has been spent as a professor of education, dean, and director of educational research centres. He is the founding director of the Walter E. Dennis Learning Center at The Ohio State University, where he directed a campus-wide centre for teaching students in academic difficulty and researched the links between motivation and school achievement (Tuckman 2003). A lengthy list of publications includes books on conducting educational research, evaluating instructional programmes, and educational psychology theory. He was Executive Editor of the *Journal of Experimental Education* and a contributing editor for a number of other journals in career development and educational research. In 2004–2005 he was an elected member of the Council of Representatives of the American Psychological Association (Tuckman 2008).

As an undergraduate psychology major at Rensselaer Polytechnic Institute in Troy, New York, Tuckman focused his senior honors thesis on instruction and learning. He later observed that 'this reflected my strong practical interest in how

people learned real information in real settings, not nonsense syllables in a laboratory' (Tuckman 1996, 3). This interest in the psychology of human learning led to graduate work at Princeton University under the mentorship of the influential learning psychologist Robert Gagné. In 1963 Tuckman obtained a Ph.D. in Psychology from Princeton. His dissertation study was published in 1964 under the title 'Personality Structure, Group Composition, and Group Functioning'. The study was supported by the Office of Naval Research and was initially presented at the 1963 meeting of the Eastern Psychological Association. It was designed to examine whether individual personality traits of group members influenced group functioning. This study reflects Tuckman's interest in group development and specifically recommends further research into 'the development of emergent group structures' (Tuckman 1964, 487).

After completing his graduate work, Tuckman took his first professional job as a research psychologist at the Naval Medical Research Institute in Bethesda, Maryland where he researched small group and organizational behaviour. In the next few years he continued exploring group dynamics and created his model of small group development. The research was driven by the Navy's need to learn about small group behaviour in light of an expected future of small crew vessels and stations. His supervisor at the time had gathered an extensive collection of articles on group development, and asked Tuckman to review them. Tuckman analyzed and interpreted the articles in terms of interpersonal and task functions of groups and subsequently categorized the stages of group development that were described in the articles.

The study was undertaken for practical reasons and, in fact, publishing it in an academic journal was not an easy task. The paper was initially turned down by the *Psychological Bulletin* on the grounds that the articles cited in the literature review were not 'of sufficient quality to merit publication'. Tuckman persevered, rewriting the document and making the case that it was not focused on the cited articles but rather used them to draw inferences about groups (Tuckman 1984). The article was published in 1965 and quickly generated a large number of reprint requests.

Overview of the model

In 1965, Bruce W. Tuckman published a literature review-based article entitled *Developmental Sequence in Small Groups*. The article focused on two realms of group development: interpersonal relationships and task activity. He hypothesized a four-stage model in which each stage needed to be successfully navigated in order to reach effective group functioning. Not until the article's summary did he coin the labels 'forming, storming, norming, and performing', which, as he later observed, 'would come to be used to describe developing groups for the next 20 years and which probably account for the paper's popularity' (Tuckman 1984, 14).

Forming

The first stage of the model is 'testing and dependence'. In this stage, the group becomes oriented to the task, creates ground rules, and tests the boundaries for interpersonal and task behaviours. This is also the stage in which group members establish relationships with leaders, organizational standards, and each other.

Storming

The second stage represents a time of intergroup conflict. This phase is characterized by lack of unity and polarization around interpersonal issues. Group members resist moving into unknown areas of interpersonal relations and seek to retain security. Tuckman (1965, 386) stated that 'group members become hostile toward one another and toward a therapist or trainer as a means of expressing their individuality and resisting the formation of group structure'. In this stage, members may have an emotional response to the task, especially when goals are associated with self-understanding and self-change. Emotional responses may be less visible in groups working toward impersonal and intellectual tasks, but resistance may still be present.

Norming

During the third phase, the group develops cohesion. Group members accept each other's idiosyncrasies and express personal opinions. Roles and norms are established. Neuman and Wright (1999) described this as a stage of developing shared mental models and discovering the most effective ways to work with each other. Tuckman (1965) stated that in this stage, the group becomes an entity as members develop in-group feeling and seek to maintain and perpetuate the group. Task conflicts are avoided in an effort to insure harmony.

Performing

In the final stage of the original model, the group develops 'functional role relatedness' (Tuckman 1965, 387). The group is a 'problem-solving instrument' as members adapt and play roles that will enhance the task activities. Structure is supportive of task performance. Roles become flexible and functional, and group energy is channelled into the task.

Adjourning

In 1977 Tuckman and Jensen were invited by *Group and Organizational Studies* to publish an update of the model. At the time Tuckman was director of educational research at Rutgers University and Mary Ann Conover Jensen was a doctoral candidate with a background in counselling psychology. Together they revisited the original model and reviewed the subsequent literature on team development. Based on this review they identified a fifth stage, 'adjourning'. This revision reflected a group life cycle model in which separation is an important issue throughout the life of the group. The revised model is shown in Figure 1.



Figure 1. Tuckman and Jensen (1977) revised model of small group development.

Limitations of the model

Tuckman (1965) identified several limitations to his original model of group development. The first, and most significant, is that the literature review did not represent a representative sample of settings where small group development processes are likely to occur. Certain settings, particularly the therapy-group setting, were significantly overrepresented. This limitation has been addressed to a limited extent by further research, but it remains largely unacknowledged that the model has been generalized well beyond its original framework.

Cassidy conducted a study to explore the extent to which Tuckman's (1965) model adequately describes group development outside of the therapy context. She studied a sample of group development models published between 1990 and 2001 and found that they all fit into a five-stage framework. However, she found a variation in the location and definition of the conflict stage. She outlined a shift in focus from looking at conflict as a stage to exploring the concerns that drive the conflict, proposing that such a shift would 'integrate the seemingly diverse models found in practitioner literature'. Cassidy also proposed that 'Tuckman's "storming" stage may not be a clearly defined stage for practitioners outside of therapeutic groups – thus limiting the applicability of Tuckman's model in experiential education' (2007, 416).

Other limitations identified by Tuckman (1965) include a lack of quantitative research rigour to his observations, and a concern with the description and control of independent variables. The model was based on a literature review and observation of a limited number of small group settings. Because of the nature of therapy groups, no attempt was made to establish controls. He stated that no conclusions about specific effects of independent variables on group development were drawn and encouraged further research along those lines.

Additional limitations have been identified through further analysis of the model. Rickards and Moger (2000) noted that the model lacks a complete explanation of how groups change over time. In addition, they identified two significant concerns relating to task performance. The first is that the model fails to address the effects of team development on creativity in problem solving. The second concern is that the model does not discuss either failure to achieve success in task performance or the ability to show outstanding performance. They ask two significant questions: first, what if the storm stage never ends, and second, what is needed to exceed performance norms?

Miller (2003) defined group development research as the investigation of group activities and how those activities change over the life of a group. After analyzing hierarchical models such as Tuckman's, she concluded that there is a high degree of consistency and similarity in the description of the stages. However, she also noted a significant number of theorists who suggest that development processes are considerably more complex than can be reflected in linear models. Sundstrom, De Meuse, and Futrell (1990, 128) echoed that concern and stated that such research may 'call into question our long-standing assumptions that the small group represents a single entity and that one model can fit all groups'.

Gersick (1988, 11) confirmed the similarity of stage-based models, noting that they are 'deeply grounded in the paradigm of group development as an inevitable progression ... researchers construe development as a movement in a forward direction and expect every group to follow the same historical path'. She identified several key criticisms of this viewpoint, including theories advocating multiple possible sequences or iterative cycles of group development. In addition, researchers

have questioned whether such models adequately address mechanisms for change over a group's lifespan, or when and how a group moves from one stage to the next. Finally, she noted that the models are limited because they frame groups as closed systems rather than addressing outside influences on group development.

Group development research and practice in the United States

Historical context

Tuckman was a psychologist, and his research came mainly from a psychotherapy and mental health context. In presenting his literature review, he divided the articles into three types: those describing therapy groups; those describing human relations training or T-groups (an intervention designed to increase interpersonal sensitivity and help people interact in more productive ways); and a third category combining articles on naturally existing groups and those on groups specifically formed to allow research on some aspect of group phenomena. The study was heavily weighted toward the therapy groups and Tuckman noted in particular that 'the dearth of group development studies in the industrial area is notable' (1965, 385).

The psychosocial framework was standard for group development research at the time. Gersick reported that research on group dynamics began in the late 1940s. She observed that the initial studies primarily focused on therapy groups, T-groups, and self-study groups, viewing a group's task in terms of personal and interpersonal goals like insight, learning, or honest communication. Tuckman's contribution, according to Gersick, was to synthesize this literature into 'a model of group development as a unitary sequence that is frequently cited today' (1988, 10).

However, even though industrial and organizational scholars were not yet studying small group development, the 1960s and 70s saw a significant change in how work was being done. Over the next few decades, research into the development cycles of small groups gained in prominence. This section will provide a brief historical review of the organizational research and show the ways the small group development model was used to inform both practice and theory.

1960s

Prior to the 1960s, organizational research had focused mainly on individual productivity. Important examples include Taylor's 1911 'time and motion studies' and Roethlisberger and Dickson's 1939 Hawthorne studies. Sundstrom, De Meuse, and Futrell pointed out that 'ever since the Hawthorne studies linked performance with group norms, their importance for work groups has been obvious but elusive' (1990, 127). In the 1940s, Lewin's research on participative management yielded theories that had implications for group decision making in the workplace. While the research was not focused on work teams, it identified the importance of involving people in decisions that affect them (Weisbord 2004).

1970s and 1980s

Tuckman's model in practice

The 1970s and 1980s saw an increased focus on team development and team building in an effort to improve interpersonal processes and productivity in the workplace. Drucker stated that there was no longer one 'right' principle of management. He

identified a new focus on teams, decentralization, and systems management, all of which led him to conclude that ‘teams have become very popular and are indeed in danger of being damaged by becoming fashionable’ (1973, 564). By the 1980s, quality circles and employee involvement groups had become common in the workplace (Sundstrom et al. 1990).

During this time the model was widely used in a variety of workplace settings. Tuckman (1984) noted that he had over 450 requests for the article during the first three or four years after it was published. Requests for permission to use the model came from across the globe and from a variety of disciplines. While he observed that the quotability of the naming scheme probably accounted for the article’s success (Tuckman 1984), an additional, and more significant, factor was the limited quantity of literature on work teams.

Tuckman’s (1965) model was frequently used by field practitioners. It was a common model in startup training for quality improvement teams. For example, Scholtes (1988) used the four stages in his team handbook, calling them ‘fairly predictable stages’ that a team goes through as it matures and members gradually learn to cope with the emotional and group pressures they face.

Tuckman’s model in theory

Tuckman and Jensen’s (1977) updated literature review identified one research study which had tested the group development model and hypothesis. In that study Runkel and others (1973) tested the four stages in a classroom setting. They found that the model was supported in that setting and encouraged others to test the hypothesis in work groups. Tuckman and Jensen identified a number of other articles that addressed group development, the vast majority of which were coming from the fields of psychology and behavioural science.

In a second study, Heinen and Jacobson (1976) examined literature on group development and proposed a model for task group development that relied heavily on Tuckman’s (1965) work. They identified several constraints, including the difficulty in determining discrete stages for group development and problems with the sequential presentation of the stages. However, they concluded that groups do appear to emerge, develop and grow in an orderly and predictable manner. Gersick (1988, 10) discussed Tuckman (1965) and Tuckman and Jensen (1977) in a historical review of group development literature and noted that ‘models offered subsequently have also kept the same pattern’.

1990s to the present

Tuckman’s model in practice

By the late 1990s the trend toward work groups and teams was well established. Cohen and Bailey (1997) noted that both the management and academic press were emphasizing the importance of teams for organizational success. Offerman and Spiros (2001) reported on several studies indicating that a significant majority of large companies were using team structures. Miller (2003, 121) stated that ‘more than ever before, organizations are recognizing the types of situations for which group work can provide a key competitive advantage’.

This trend had significant impact for organization development practitioners. On the one hand, they were facing an increased demand for services to promote

effectiveness of work groups. On the other hand, the research literature was limited, particularly in terms of practical application. Offermann and Spiros (2001, 376) observed that ‘the increasing organizational reliance on teams, coupled with a literature criticized for limited utility to real-world problems, is pushing a practice in which practitioners allegedly favour shotgun approaches that combine multiple intervention strategies in the hope that something will work’.

Tuckman’s (1965) model was a useful starting point for team development practitioners. Because the model was accessible, easy to understand, and flexible enough to apply to many different settings, it was frequently mentioned in management and practitioner literature (Nash and Bolin 2003; Parker 1990; Robbins and Finley 2000). In a survey of 150 professionals, Offerman and Spiros (2001) identified 250 different models and theories that were being used in team development practice. Of these, Tuckman’s model was the most common, mentioned by 16% of respondents.

Tuckman’s model in theory

During the 1990s there was a significant increase in the academic literature on group development. Wheelan (1997, 288) observed that ‘recent interest in group productivity in the workplace has led to a plethora of research on group processes and development’. By this time the question of the model’s generalizability was moot. The model began to appear frequently in the scholarly literature. It was regularly listed as a reference on group development theory, and was being widely applied as a basis for common understanding in research on work groups. Researchers were regularly citing Tuckman (1965) and Tuckman and Jensen (1977) in discussions of team development. Many were responding to the model, building upon it, or refuting it. For example, Levin (1992) cited both the original and revised models in a discussion of quality improvement teams. Miller (2003) used Tuckman’s (1965) model in a study of team processes, stating that her research would be most effective if it started from Tuckman’s ‘well-established paradigm’.

At the beginning of the twenty-first century, the model was being applied in studies of a wide variety of work settings from project teams (Erickson and Dyer 2004; Rickards and Moger 2000) to leadership teams (Wheelan 2003) and even public health partnerships (McMorris, Gottlieb, and Sneden 2005). As office technology broadened in importance, the model was applied to development of virtual teams (Furst, Blackburn, and Rosen 1999; Maruping and Agarwal 2004). In addition, Cassidy (2007) noted that within education it remained one of the most commonly cited group development models.

Implications

Group development has been an important area of HRD research for the past 50 years. The Tuckman (1965) and Tuckman and Jensen (1977) model of small group development has a unique history, in that it has been widely referenced by both academic researchers and HRD practitioners. The model created a starting point for conversations between the academy and the field. It provided a needed baseline of agreement on terms and ideas. McMorris, Gottlieb, and Sneden stated that ‘one of the strengths of the Tuckman model is its ease of use at the practitioner level’, noting its practical perspective and commonsense approach. At the same time, they chose it

as the basis for their study on developmental stages in public health partnerships because it provided a framework and an effective lens for viewing practice settings (2005, 291).

It is, perhaps, unlikely that a model with similar impact will come out of the new literature. First, recent theories recognize the complexity of group dynamics in today's world and are not easily represented in a simple model. Second, the wide body of literature on organizational and workplace issues means that practitioners have access to information about many specialized areas of group development such as leadership, motivation and rewards. These theories are exponentially broader and deeper than Tuckman's original model. They provide detailed discussion of many aspects of group dynamics from forming through adjourning. They also examine external factors affecting group development, including organizational roles, resource allocation, and pressure from external stakeholders. They do not, however, provide the same breadth of application. HRD scholars and practitioners can learn something from a model that has proved valuable for almost 45 years. The utility of providing a simple, accessible starting point for conversations about key issues of group dynamics has not diminished.

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